Md Hasebul Hasan

Ph.D. Student in Computer Science and Engineering | Trustworthy, Human-Centered, and Efficient AI

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Research Interests

Human–AI interaction, natural language processing, trustworthy AI, LLM evaluation and safety, agentic systems, and efficient LLM inference. I develop methods and benchmarks that diagnose model failures, improve reliability in real-world interactions, and reduce the computational cost of deployment.

Education

Ph.D. in Computer Science and Engineering **Aug 2025 – Present**
University of Texas at Arlington, Arlington, TX *GPA: 4.00/4.00*
Advisor: Mohammad Atiqul Islam. Graduate Research Assistant.
Expected graduation: June 2028.

M.Sc. in Computer Science and Engineering **Jul 2022 – Aug 2025**
Bangladesh University of Engineering and Technology (BUET), Dhaka *GPA: 3.75/4.00*
Thesis: *An Efficient Compositional Reasoning Approach for Solving Geospatial Queries Using LLMs.* Research led to publications at ICML, EACL, and IEEE COMPSAC.

B.Sc. in Computer Science and Engineering **Feb 2017 – May 2022**
Bangladesh University of Engineering and Technology (BUET), Dhaka *GPA: 3.77/4.00; Rank: 31/127*
Senior-year GPA: 3.89/4.00. Thesis: *A Multi-Modal Deep Learning Based Approach for House Price Prediction.*

Research Experience

Graduate Research Assistant **Aug 2025 – Present**
University of Texas at Arlington *Advisor: Mohammad Atiqul Islam*

- Research trustworthy, human-centered, and efficient LLM systems, with emphasis on real-world evaluation, small-language-model adaptation, and memory-efficient inference.
- Created **DeEscalWild**, a benchmark derived from real-world police–civilian interactions containing **1,500 scenarios, 285,887 dialogue turns, and approximately 4.7M tokens**; designed the transcription, diarization, filtering, anonymization, validation, and evaluation pipeline.
- Designed controlled evaluations across multiple 2B–3B language models, including fine-tuning, automatic metrics, LLM-as-a-judge evaluation, expert human review, and latency measurement.
- Investigate KV-cache compression for Llama-2-7B on NVIDIA H100 GPUs, studying low-rank projection, token retention, oracle allocation, K/V asymmetry, memory, throughput, and latency trade-offs.

Graduate Researcher – GeoAI and LLM Agents **2022 – 2025**
Bangladesh University of Engineering and Technology *M.Sc. research*

- Co-developed **MapEval**, a benchmark for geospatial reasoning in foundation models; led literature-gap analysis and helped design the spatial reasoning taxonomy, dataset construction, annotation, and human verification.
- Led **MapAgent**, a hierarchical agent with dynamic map-tool integration for compositional geospatial reasoning; first author on the EACL 2026 Findings paper.
- Developed **GuardFL**, a majority-consensus approach for detecting and removing malicious clients in federated learning; first author on the IEEE COMPSAC 2024 paper.

Publications

Peer-Reviewed / Accepted

1. **M. H. Hasan**, K. H. Charu, E. P. Sridhar, S. Deb, and M. A. Islam. "DeEscalWild: A Real-World Benchmark for Automated De-Escalation Training with SLMs." *NeurIPS 2026 Evaluations and Datasets Track*, 2026. **Accepted**. First author; led benchmark construction, data validation, model fine-tuning, and evaluation. [paper]
2. M. L. Dihan, M. T. Hassan, M. T. Parvez, **M. H. Hasan**, M. A. Alam, M. A. Cheema, M. E. Ali, and M. R. Parvez. "MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models." *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. **Spotlight (top 2.6%)**. [paper]
3. **M. H. Hasan**, M. L. Dihan, T. Hashem, M. E. Ali, and M. R. Parvez. "MapAgent: A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration." *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026. [preprint]

4. **M. H. Hasan**, M. S. I. Khan, M. S. Haque, A. Y. Ahmed, and M. A. Adnan. "GuardFL: Detection and Removal of Malicious Clients Through Majority Consensus." *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 1112–1117, 2024. [paper]

Manuscripts / Preprints

5. **M. H. Hasan**, M. A. Jahan, M. E. Ali, Y.-F. Li, and T. Sellis. "A Multi-Modal Deep Learning Based Approach for House Price Prediction." arXiv:2409.05335, 2024. First author; led multimodal architecture design and evaluation. [preprint]

Industry Experience

Machine Learning Engineer **May 2022 – Jul 2025**
IQVIA, Dhaka, Bangladesh *Full-time*

- Built and deployed ML prediction systems using Python, PyTorch, TensorFlow, and Snowflake; production work contributed to solutions associated with a 15% increase in sales.
- Developed time-series data pipelines over enterprise data marts, including data engineering, preprocessing, model optimization, evaluation, and deployment.
- Architected LangChain-based multi-agent workflows, fine-tuned open-source LLMs for domain-specific applications, and conducted AI security red-team assessments.
- Implemented scalable MLOps with AWS SageMaker/Bedrock, Kubernetes, Kubeflow, Docker, ECR/EC2, Git, and GitLab CI/CD.

Teaching Experience

Lecturer, Department of CSE **Nov 2022 – Jul 2024**
Bangladesh University of Engineering and Technology (BUET) *Part-time*

- Taught six undergraduate sessional/laboratory courses: Algorithm Engineering, Software Development, Databases, Data Structures and Algorithms II, Digital Logic Design, and Structured Programming.
- Supervised **30 student projects**; designed and graded assignments, laboratory exercises, and examinations for classes of 40–60 students.

Research and Travel Grants

- **NeurIPS 2026 Financial Assistance Award.** Covers accommodation and registration for both the conference and workshops in Atlanta, GA.
- **Travel Grant, The University of Texas at Arlington, 2026. US$500** to present *DeEscalWild* at NeurIPS 2026 in Atlanta, GA.
- **Research Grant for M.Sc. Thesis**, Directorate of Advisory, Extension and Research Services, BUET, 2024. Competitive award of approximately **US$1,500** supporting MapAgent research.

Awards & Honors

- **ICPC South Central Regional, Division 2 Champion**, November 7, 2026. *Code Blazers*, The University of Texas at Arlington; solved **9 of 13 problems**. Team: Nguyen Le, Md Hasebul Hasan, Nghia Nguyen Phuoc Huu; coach: Donna French.
- Platinum Award, IQVIA Impact Program, 2024; Silver Award, IQVIA Impact Program, 2025.
- University Merit List Scholarship (two semesters).
- Dean's List for three consecutive years (Levels 2–4), BUET.

Technical Skills

Programming: Python, C/C++, Java, JavaScript, SQL, LaTeX
ML / AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, LLM fine-tuning, RAG, agentic systems, model evaluation
Cloud / Systems: AWS SageMaker, Bedrock, EC2, ECR, Docker, Kubernetes, Kubeflow, Git/GitLab CI/CD, NVIDIA GPU/HPC workflows
Databases / Web: Snowflake, PostgreSQL, MySQL, MongoDB; Node.js, Vue.js, React

References

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